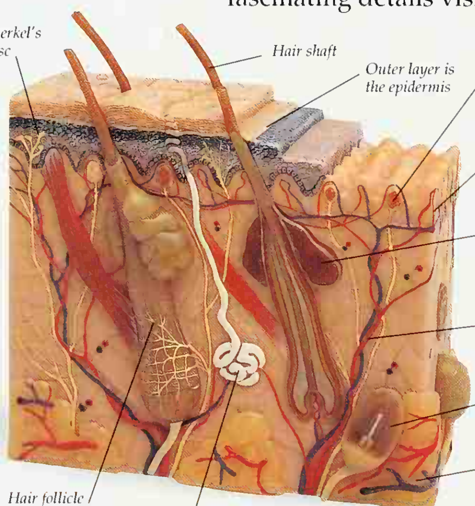
Skin and touch

ARISTOTLE OF ANCIENT GREECE listed the five main senses as sight, hearing, smell, taste, and touch. Unlike the other sense organs, the body part involved in touch is not concerned with sense alone. It has many other functions. Indeed, it is the body's biggest organ – the skin. On an adult, this living outer covering weighs about 11 lb (5 kg) and has an area of some 2.4 sq yd (2 sq m). Its tough surface layer, the epidermis, continually replaces itself to repair wear and tear, and to keep out water, dust, germs, and harmful ultraviolet rays from the sun. Under this is a thicker layer, the dermis, which is packed with nerves, blood vessels, and stringy fibers of the body proteins, collagen and elastin. The dermis also assists in temperature regulation by sweating (p. 39), and by turning pale or flushed. Skin was ignored by many great anatomists. They dismissed it as something to be removed in order to study the more interesting parts beneath. Like many other body parts, the invention of the microscope made the skin's fascinating details visible.

FINGERTIP READING
Braille and other touch-reading systems help people with limited eyesight. Fingertip skin is sensitive enough to detect patterns of raised dots representing numbers and letters. French inventor Louis Braille (1809-1852), was blinded at three years of age, and began to devise his system at the age of 15.

FRICTION RIDGES
The epidermal skin layer of the fingers, palms, soles, and toes is folded and tucked into swirling patterns of ridges. Each ridge is about 0.008-0.016 in (0.2-0.4 mm) wide. Along with the slight dampness from sweat and oil-producing sebaceous glands, the ridges help the skin to grip without slipping. On the fingers, these are called fingerprints. They are classified into types by the presence of three main features: arches, loops, and whorls. Each human has its own unique set of fingerprints.



UNDER THE SKIN
The surface of the skin is dead. It consists of flat, interlocking dead cells filled with the hard-wearing protein keratin. These wear away and are replaced by cells moving up from below, like a conveyor belt. The cells are produced by continual division (p. 48) at the base of the skin's upper layer, the epidermis. The dermis is much thicker and contains various microscopic sensors responsible for touch – which is a combination of light pressure, heavy pressure, heat, cold, and pain. The dermis houses about 3 million tiny coiled sweat glands (p. 39), and roughly as many hair follicles, from which hairs grow.

Light micrograph of skin, showing presence of melanin

DEAD HAIRS
In this scanning electron micrograph, hairs protrude from their pitlike follicles in the skin. Only the base of the hair, where it grows, is alive. The shaft that shows above the surface is, like the skin around it, made chiefly of keratin and quite dead. There are some 100,000 hairs on the average head. Each head hair grows at the rate of 0.04 in (1 mm) every three or four days. There are also much tinier hairs over the rest of the body. The only hairless skin is on the palms of the hands, the palm sides of the fingers, and the soles of the feet.

SKIN COLOR DIFFERENCES
One of the differences between humans and other primates is the much smaller, finer hairs covering the body, so that humans look almost hairless compared to monkeys and apes. Thus, humans have less protection from the Sun's ultraviolet rays, which can damage the skin and the tissues underneath. The presence of melanin, a dark brown pigment made by cells in the skin, protects the tissues beneath from the Sun's rays. As humans moved to more temperate regions, having lots of pigment became less important.

Sun-damaged skin, due to lack of melanin